Pore size optimization for triboelectric performance

"What pore size range maximizes triboelectric output in SiO2-MoS2 composites?" *Generated by LeapSpace*

Best-supported answer

No exact pore-size range has been established for maximizing triboelectric output in *SiO_2–MoS_2* composites specifically. The closest TENG evidence supports **nanoporous to mesoporous structures**, with **~20–40 nm pores in SiO_2** giving higher triboelectric potential as porosity increased, while other porous TENG materials show best performance at **medium porosity rather than the largest pores** [46] [75] [92].

Evidence type	What it shows	Relevance to SiO_2–MoS_2 TENGs
Direct TENG evidence on porous SiO_2	Rice-husk SiO_2 with **20–40 nm pore size** and more porosity produced much higher triboelectric potential, rising from **~253 V** for flat SiO_2/PTFE to **2336 V** with increased porosity [92].	This is the strongest directly relevant pore-size evidence for the SiO_2 side of your system.
Direct TENG evidence on porous silica composites	A porous silica/PDMS TENG using **multi-sized pores** showed the **medium-porosity** sample performed best, reaching **4.37 W/m²** and outperforming pure PDMS by **6.52×** [46].	Suggests that an intermediate pore architecture is better than simply maximizing pore size or void fraction.
General porous-TENG design evidence	Porosity increases triboelectric output by enlarging contact area and creating more charge-trapping sites [48] [50] [65].	Supports using a porous or mesoporous structure in the SiO_2–MoS_2 composite.
MoS_2/SiO_2 TENG mechanism evidence	SiO_2 contributes high charge capacity, while the MoS_2/SiO_2 interface supports high charge density and slow charge decay [11] [34].	Means pore engineering should preserve shell continuity and interfacial charge trapping, not just maximize pore size.

Practical takeaway for SiO_2–MoS_2 composites

- The most defensible target is **small-to-moderate mesoporosity / nanoporous structure**, not large macropores [46] [75] [92].
- **Very large pores are not shown to be optimal** in the retrieved TENG evidence; instead, **intermediate porosity** gives the best balance of contact area, charge trapping, and structural integrity [46] [75].

- For **SiO_2–MoS_2**, the pore architecture should likely be tuned to **increase effective surface area while keeping the MoS_2 shell continuous**, because the shell's humidity shielding and charge retention are part of the performance gain [11] [34].

Bottom line

If you want a single working target from the retrieved evidence, aim for **mesoporous/nanoporous pores in the tens-of-nanometers scale, with intermediate porosity**, because that is where the clearest TENG gains appear in silica-based systems [46] [92]. **A precise optimum for SiO_2–MoS_2 composites remains unreported in the retrieved sources**, so any exact "best pore size" beyond that would be an inference [11] [34].

Confidence level: **Medium** — the evidence is strong for porous silica-based TENG enhancement and for the importance of preserving the SiO_2–MoS_2 interface, but it does not contain a direct pore-size sweep for the exact SiO_2–MoS_2 composite.

References

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